WORKSHOP SERIES 2025

AI FOUNDATIONS

Prompt Engineering & Applied AI

A beginner-friendly workshop for Business Students

Understanding AI · ML · DL · LLMs · Generative AI

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Workshop Agenda

What we'll cover today

01

What is AI?

Definition, history & real-world examples

02

AI vs ML vs DL

Understanding the hierarchy & differences

03

LLMs & Generative AI

How language models work & why they matter

04

Business Applications

AI tools transforming industries today

05

Prompt Engineering

Writing smart prompts for better AI results

06

Hands-on Practice

Apply AI in your business field

What is Artificial Intelligence?

"Artificial Intelligence is the simulation of human intelligence by machines — enabling computers to think, learn, reason, and solve problems."

🧠 Think

Process information and draw conclusions like humans

📚 Learn

Improve performance from data & experience

🗣️ Communicate

Understand language & respond naturally

⚡ Automate

Perform repetitive tasks faster and smarter

💡 **Business Insight: AI is your smartest digital assistant — available 24/7, never tired, always learning!**

A Brief History of AI

From science fiction to everyday reality

1950s

Alan Turing proposes the 'Turing Test'. AI as a field is born.

1980s

Expert Systems emerge. Computers follow human-coded rules to solve problems.

2000s

Machine Learning takes off. Algorithms learn from data automatically.

2012

Deep Learning revolution. Neural networks begin beating human performance.

2017

Transformer architecture invented — the foundation of modern AI.

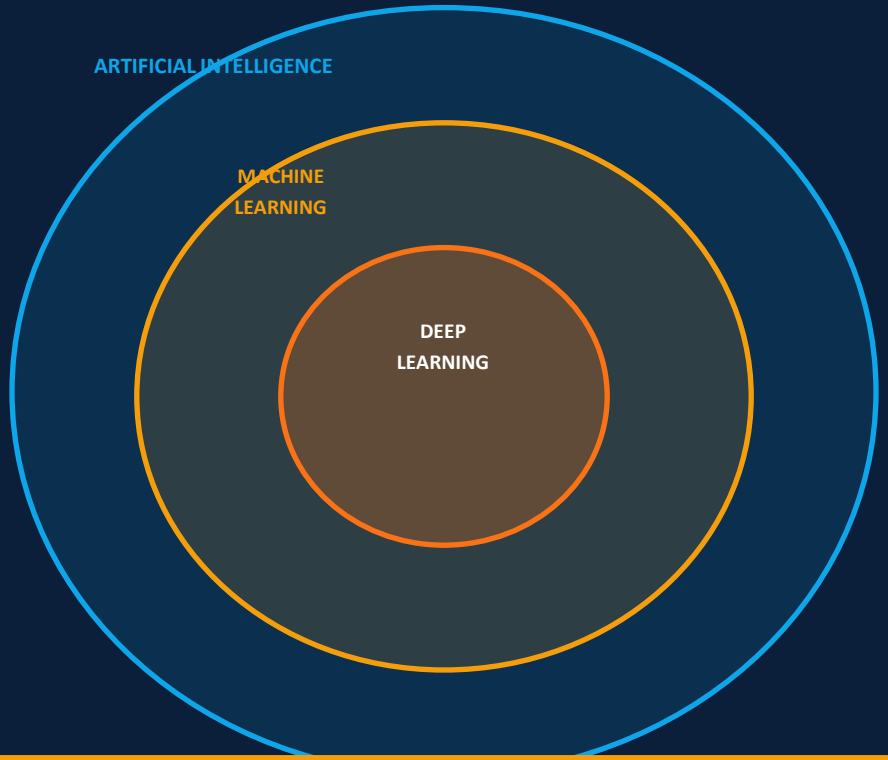
2022+

ChatGPT & Generative AI explode. AI enters every business & home.

Today, AI is no longer the future — it IS the present. Every business student must understand it!

AI · ML · DL

Understanding the Hierarchy — The Most Important Distinction in AI



AI — Artificial Intelligence

The Broadest Concept

Any technique that enables machines to mimic human intelligence. Includes rule-based systems, expert systems, ML, and DL.

ML — Machine Learning

Subset of AI

Algorithms that LEARN from data automatically — without being explicitly programmed for every rule. It finds patterns.

DL — Deep Learning

Subset of ML

Uses deep neural networks inspired by the human brain. Powers image recognition, speech, and language understanding.

🎯 **Key Takeaway: Every DL is ML, every ML is AI — but NOT every AI is ML or DL!**

AI vs ML vs DL — Side-by-Side Comparison

A clear breakdown for business students

Feature	🤖 Artificial Intelligence	📊 Machine Learning	🧠 Deep Learning
Definition	Machines mimicking human intelligence	Algorithms learning from data	Neural networks with many layers
Data Needed	Can be rule-based, less data needed	Needs structured data	Needs LARGE amounts of data
How It Works	Follows explicit rules or learns	Finds patterns in data	Learns complex patterns (images, text)
Complexity	Broad & varies	Medium	High — computationally intensive
Business Use	Chatbots, automation	Recommendations, predictions	Image recognition, language AI
Examples	Siri, Chess Engine	Netflix, Amazon suggest	ChatGPT, Google Translate, DALL-E

Large Language Models (LLMs)

The engine behind ChatGPT, Gemini, Claude & modern AI tools

An LLM is an AI model trained on billions of words of text — books, websites, code, articles — to understand and generate human language. It predicts the next word based on context, producing fluent, meaningful responses.

1

TRAINING

Fed trillions of words from internet, books, code

2

LEARNING

Learns grammar, facts, reasoning, and context

3

PREDICTION

Predicts the most likely next word/token

4

GENERATION

Produces full sentences, essays, code, or answers

Popular LLMs: GPT-4 (OpenAI) · Gemini (Google) · Claude (Anthropic) · Llama (Meta) · Mistral

Generative AI — What Can It Create?

AI that doesn't just analyze — it CREATES

Generative AI refers to AI systems that can generate NEW content — text, images, audio, video, code — that didn't exist before. It learns patterns from training data and creates original output based on your instructions (prompts).

✍️ Text & Writing

- Reports & Emails
- Marketing copy
- Stories & scripts

🖼️ Images & Art

- Logos & designs
- Product mockups
- Presentations

💻 Code

- Write programs
- Fix bugs
- Build websites

🎵 Audio & Video

- Music generation
- Voice cloning
- Video synthesis

🌟 **GenAI Tools You Know: ChatGPT · DALL-E · Midjourney · Sora · Copilot · Gemini · Claude**

How Do LLMs Actually Work?

A simple, non-technical explanation

YOU TYPE a Prompt

"Write a business plan for a café"

→

TOKENIZATION (Words → Numbers)

Each word split into small units called tokens

→

TRANSFORMER Processing

Self-attention weighs relationships between all words

→

AI GENERATES Response

Predicts word-by-word, building the perfect answer

Key Technical Concepts — Simply Explained:

Tokens: Smallest units of text (words or word-parts) the AI processes

Parameters: Billions of learnable numbers that store knowledge (GPT-4 has ~1.8 trillion!)

Attention: The AI's ability to focus on relevant words in context

Temperature: Controls creativity — low = factual, high = creative & varied

Why AI Matters for Business Students

The competitive edge you can't afford to ignore

85M+

Jobs transformed by AI by 2025

$15.7T

AI contribution to global GDP by 2030

97%

of business leaders using AI tools

How AI Is Used Across Business Sectors:

Marketing

Personalized ads, content generation, customer segmentation, chatbots

Finance

Fraud detection, credit scoring, algorithmic trading, risk analysis

Healthcare

Disease diagnosis, drug discovery, patient monitoring, medical imaging

E-Commerce

Product recommendations, demand forecasting, customer service bots

Management

Decision support, process automation, HR analytics, strategy planning

Education

Personalized learning, auto-grading, tutoring bots, content creation

Key Features of Modern AI & LLMs

What makes today's AI so powerful and versatile?

Multilingual

Understands and generates content in 100+ languages simultaneously

Multi-Modal

Processes text, images, audio, and video all at once

Conversational

Maintains context across long conversations, remembers what you said

Real-Time Speed

Generates thousands of words per second — instantly

Customizable

Fine-tuned for specific industries, roles, or company needs

Scalable

Serves millions of users simultaneously without quality drop

Prompt Engineering — Speak AI's Language

The skill that separates good from great AI results

A PROMPT is the instruction/question you give to an AI. Better prompts = dramatically better results.

❌ WEAK PROMPT

"Write about marketing"

Result: Vague, generic content that could be for anyone. Not useful.

✅ STRONG PROMPT

"Act as a marketing consultant. Write 3 social media strategies for a Kathmandu-based café targeting young professionals aged 22–35. Include Instagram and TikTok ideas."

Result: Targeted, actionable, specific content. Very useful!

The P.R.A.C.T. Framework for Great Prompts:

P — Person/Role
"Act as a business analyst..."

R — Request
Clear, specific task

A — Audience
Who is this for?

C — Context
Background info needed

T — Tone/Format
"Give me a bullet list..."

AI in Action — Real Business Use Cases

How companies are using AI RIGHT NOW — and how you can too!

🏢 Netflix | Recommendation AI

Netflix's recommendation engine (built on ML) saves $1 billion/year by reducing churn. 80% of what people watch comes from AI suggestions.

🏢 Amazon | Demand Forecasting AI

Amazon uses ML to predict what products customers will buy before they search. They ship to warehouses BEFORE orders — cutting delivery time dramatically.

🏢 Banks & FinTech | Fraud Detection AI

Banks use real-time ML to detect fraudulent transactions in milliseconds. The system analyzes hundreds of data points simultaneously to flag suspicious activity.

🏢 ChatGPT/Copilot | Productivity AI

Businesses use LLMs to write emails, reports, code, and presentations — cutting content creation time by 60-80%. Already used by 100M+ people daily.

💼 Your Opportunity: Learn AI now and become the person your future employer needs!

AI Limitations & Ethical Considerations

Being a responsible AI user means understanding the risks

🎭 Hallucinations

AI can confidently state WRONG information. Always verify important facts from authoritative sources.

📅 Knowledge Cutoff

LLMs have a training cutoff date — they don't know about recent events unless given web access.

⚖️ Bias in AI

AI learns from human data, inheriting human biases. Outputs may reflect gender, racial, or cultural biases.

🔒 Privacy Risk

Don't share sensitive company data or personal info with public AI tools. What you share may be used for training.

📋 Copyright Issues

AI-generated content may inadvertently replicate copyrighted material. Always review before publishing.

🤖 Job Displacement

AI automates some jobs while creating new ones. Adaptability and AI literacy are your best protection.

⚠️ **Golden Rule: AI is a powerful TOOL, not an all-knowing oracle. Think critically, verify claims, use responsibly.**

Key Takeaways

What every business student should remember from today:

01 AI is a broad field; ML and DL are powerful subsets that learn from data.

02 LLMs like ChatGPT are trained on massive text data to understand and generate language.

03 Generative AI can create text, images, code, audio — revolutionizing every industry.

04 Better prompts = dramatically better AI results. Prompt engineering is a real skill.

05 Every business sector — marketing, finance, HR, management — is being transformed by AI.

06 Use AI responsibly: verify facts, protect privacy, and stay critically aware of limitations.

Thank You!

You're now equipped with AI foundations that will set you apart.

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🚀 **Your Next Steps:**

1. Try ChatGPT / Claude / Gemini today with strong prompts
2. Identify ONE task in your studies/work AI can help with
3. Follow AI news (stay curious — this field moves fast!)